

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Built and shipped features for a multi-tenant NHS eTriage API platform (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack platform for creating machine-learning training datasets (React, Node.js, MongoDB) used daily by the AI team, cutting manual data work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Built platform security features, **AWS KMS** encryption for sensitive data, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**.
- Owned the eTriage backend with full autonomy, maintained internal tooling and technical documentation, mentored a second developer to on-call readiness, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories to **TypeScript** and shipped a production LLM feature (auto-commenting via **4-bit quantized Llama-7B, llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with Stripe hybrid payments, cron automation, and **git-webhook CI/CD auto-deploy**; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service that crawls the web, filters and stores results, and turns open-ended queries into structured entity tables via typed retrieval planning, deterministic extraction with LLM fallback, and **cell-level provenance**; hardened with **150+ tests** and provider routing.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Designed a robust **PySpark** data pipeline on a **~1.4B-row** dataset with broadcast joins, adaptive execution, partitioning, column pruning, and a 6-rule data-quality filter; cutting end-to-end runtime **~25%** and tail-latency ratio **2.14x to 1.74x** for reliability.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built a scalable edge service in **TypeScript** on Cloudflare Workers with durable, restart-safe workflows and D1 storage, serving a streaming AI agent over WebSocket that survives process restarts and individual tool failures.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, SQL, C++, Java, PostgreSQL, MongoDB, SQLite

Backend & APIs: FastAPI, Node.js, Express, REST APIs, async services, integration testing

Data & Pipelines: PySpark, Spark, ETL, web crawling, data filtering, Spark UI

ML & AI: PyTorch, Hugging Face Transformers, LLM agents, RAG, sentence-transformers, evaluation

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Cloudflare Workers, Linux

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.